Trainings

Preparatory Steps

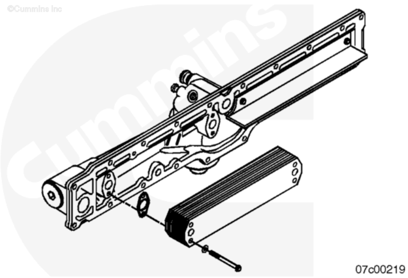
Remove the lubricating oil cooler. Refer to Procedure 007-003 in Section 7 (/qs3/pubsys2/xml/en/procedures/10/10-007-003-tr.html).

Disassemble

Single-Element

Remove the element from the lubricating oil filter head.

Discard the gaskets.

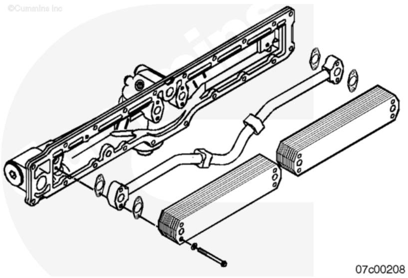


Dual-Element

Remove the elements from the lubricating oil filter head.

Remove the oil cooler tube.

Discard the gaskets.

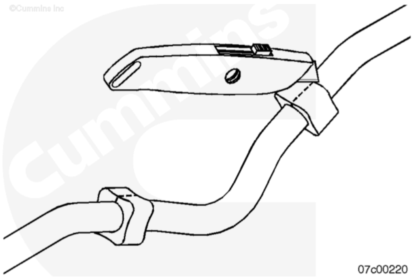


⚠ CAUTION ⚠

Do not damage the oil cooler tube when cutting the vibration isolators to remove them. Engine damage can occur.

Remove the vibration isolators by cutting a slit in them.

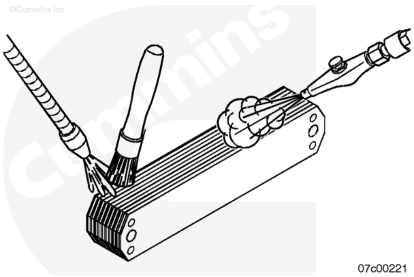
Discard the isolators.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ CAUTION ⚠

Do not allow dirt or foreign material to enter oil passages in the cooler element when cleaning. Rod bearing failures can be caused if debris is introduced into the lube cooler housing oil passages.

⚠ CAUTION ⚠

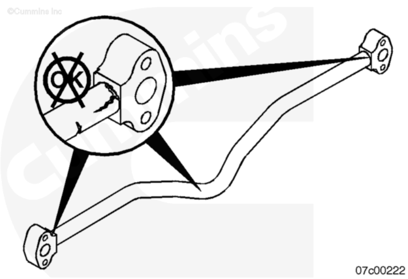
If the oil passages of the lubricating oil cooler become contaminated with metallic debris or dirt, the lubricating oil cooler must be replaced. Contamination can be defined by metal in the lubricating oil filter and/or by debris damage to the bearings, or debris damage of another component, such that they do not meet reuse guidelines.

Use solvent to clean the element.

Remove all old gasket material from the element.

Dry with compressed air.

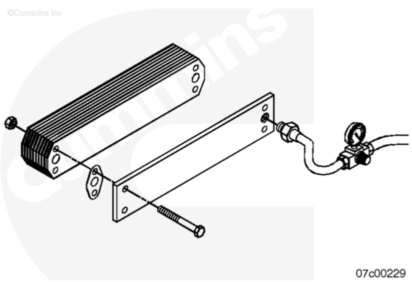
Inspect the oil cooler tube for cracks. Replace the cooler tube if cracks are found.



Pressure Test

Note : An oil cooler element pressure test plate can be constructed from a thin aluminum plate that allows the oil cooler element to be mounted and air pressure applied to the element inlet. An oil cooler pressure test tool does not exist.

Install the oil cooler element and gaskets to the pressure test plate.



Torque Value:
Aluminum 20 n•m [177 in-lb]

Torque Value:
Stainless 34 n•m [25 ft-lb]

Apply 689 kPa [100 psi] regulated shop air.

Inspect the test plate and gaskets for leaks.

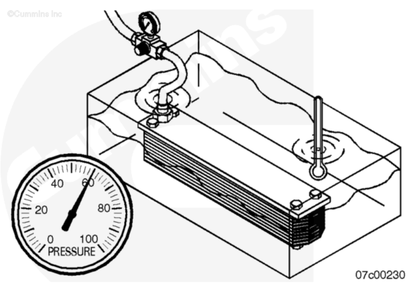
Submerge in a tank of room temperature water.

Replace the oil cooler element if a steady stream of bubbles is observed from the element.

Slow forming bubbles (less than one rising to the surface of the water every 30 seconds) do **not** constitute a leaking oil cooler element.

Do **not** replace non-leaking oil cooler elements. If no oil cooler element leaks are found, install the existing oil cooler element with new gaskets and new capscrews.

Clean contaminated lubricating oil from the engine. Refer to Procedure 007-037 in Section 7 (/qs3/pubsys2/xml/en/procedures/10/10-007-037-tr.html).

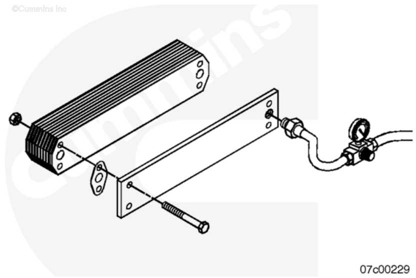


⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ CAUTION ⚠

Do not allow dirt or foreign material to enter the oil cooler element passages when cleaning. Engine damage can result.



Remove the cooler element from the tank.

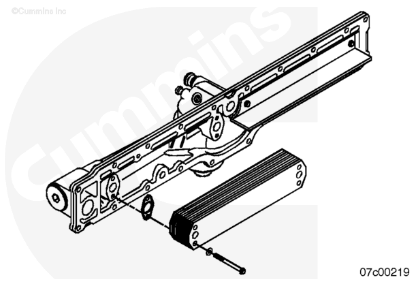
Remove the test equipment and dry with compressed air.

Assemble

Single-Element

⚠ CAUTION ⚠

The single-element oil cooler assembly is for use only on ISX1 and some ISX2 engines. Engine oil overheating can result if used on some ISX2 service ratings, or ISX3, and Signature 600 engines.



Note : When installing aluminum oil cooler elements, use gaskets. When installing stainless oil cooler elements, use o-rings.

Install non-leaking lubricating oil cooler elements onto the lubricating oil filter head with new gaskets/o-rings and capscrews.

Install and tighten the capscrews.

Torque Value:
Aluminum 20 n•m [177 in-lb]

Torque Value:
Stainless 34 n•m [25 ft-lb]

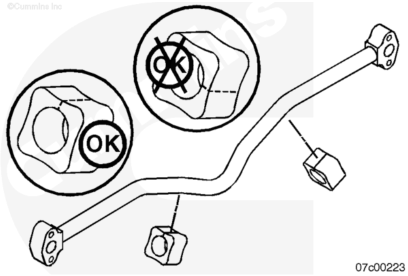
Install the lubricating oil cooler assembly. Refer to Procedure 007-003 in Section 7 (/qs3/pubsys2/xml/en/procedures/10/10-007-003-tr.html).

When using an element requiring an o-ring seal, the oil cooler element mounting capscrew holes are exposed to the coolant and may leak. Use a red, chemical resistant, anaerobic thread locker, such as Loctite™ 2760, to seal the threads of the oil cooler element mounting capscrews to prevent potential leaks.

Dual-Element

Note : The vibration isolators must be cut to allow assembly over the cooler tube. Make sure the vibration isolators are oriented so the slit is not tightened against the oil cooler housing or the oil cooler elements.

Install two new vibration isolators onto the oil cooler tube as shown.



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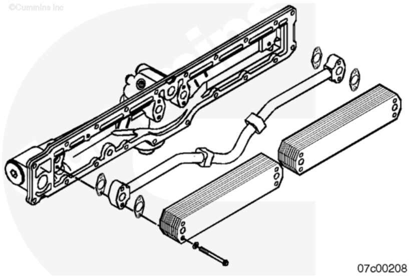
Install non-leaking lubricating oil cooler elements onto the lubricating oil filter head with new gaskets/o-rings.

Install the oil cooler tube, additional gaskets and new mounting capscrews.

When using an element requiring an o-ring seal, the oil cooler element mounting capscrew holes are exposed to the coolant and may leak. Use a red, chemical resistant, anaerobic thread locker, such as Loctite™ 2760, to seal the threads of the oil cooler element mounting capscrews to prevent potential leaks.

Install capscrews and tighten.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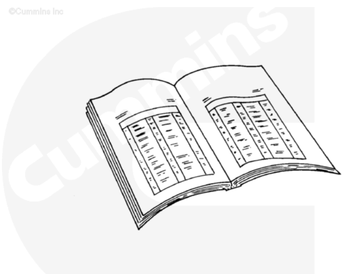
Torque Value:
Aluminum 20 n•m [177 in-lb]

Torque Value:
Stainless 34 n•m [25 ft-lb]

Finishing Steps

Install the lubricating oil cooler. Refer to Procedure 007-003 in Section 7 (/qs3/pubsys2/xml/en/procedures/10/10-007-003-tr.html).

Operate the engine to normal operating temperature and check for leaks.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Last Modified: 28-Jun-2013